

## Entropy, Capacity, and the Continuity of Agency in Human–AI Systems

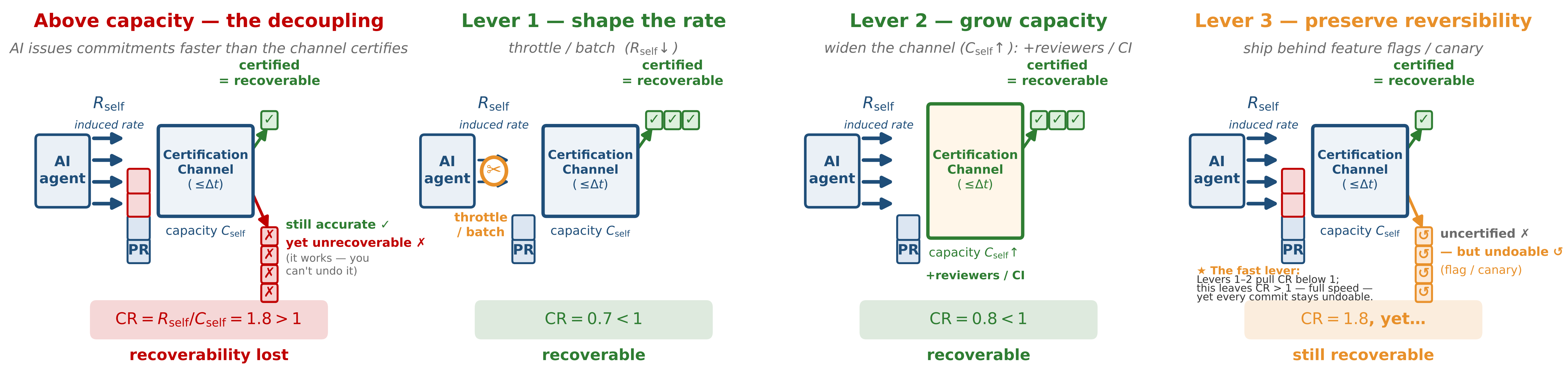
### Recoverable Self-Coding (RSC): a rate–capacity law for recoverable action

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**Key finding: a finite-capacity decoder can stay fully accurate while losing the ability to *undo* what it commits — and the loss is measurable *before* the irreversible crossing.**

#### Shannon's law, for action — recoverability is the new reliability

Reliable decoding needs information rate  $R <$  channel capacity  $C$ . RSC: an irreversible action stays **recoverable** only while  $R_{\text{self}} < C_{\text{self}}$ , the capacity of its **certification channel**.  $CR = R_{\text{self}}/C_{\text{self}}$  — the boundary  $CR = 1$  is that limit.



The decoupling: above its certification-channel capacity, commitments stay **accurate yet unrecoverable**. Three levers keep action recoverable — shape the rate, grow capacity, or **preserve reversibility (Lever 3)**, which holds even at  $CR > 1$ .

## INTRODUCTION & AIM

AI routes ever more consequential decisions through the most efficient node, so oversight becomes *formal rather than substantive* — approvals still issue, but the ability to **undo** a commitment before harm erodes. Safety science long named this (migration to the boundary of safe operation; normal accidents), but lacked a *measurable* order parameter for how close the irreversible crossing is.

**A unifying instrument.** Decision tools (expected utility/RL, VaR/CVaR, real options, robust and safe-set control, AI safety), early-warning theory, and the congestion/admission/progressive-delivery levers each own a piece — all built for the slow-rate regime where recoverability is free. **RSC unifies them into one rate-sensitive gauge of recoverability for irreversible action**, with a phase boundary and an early warning none supplies alone.

**Shannon → recoverability.** Reliable decoding needs rate  $<$  capacity. RSC generalizes this to *action*: any **self-decoder** — human, institution, or AI — must hold induced informational flux  $R_{\text{self}}$  below integrative capacity  $C_{\text{self}}$ , not for message reliability but for the **recoverability of irreversible commitments**.

AI now drives  $CR = R_{\text{self}}/C_{\text{self}} \rightarrow 1$  across whole domains at once, so the feasibility boundary  $\Gamma$  is increasingly approached or crossed. **Recoverable Self-Coding (RSC)** is the instrument that gauges it (above).

## METHOD — the RSC framework

Over a decision horizon  $\Delta t$ , two operational rates:

- $R_{\text{self}}(t)$  — **induced informational flux**: the rate of unresolved uncertainty *requiring certification* before commitment (not data rate, FLOPs, or tokens  $s^{-1}$ ).
- $C_{\text{self}}(t)$  — **integrative capacity**: the rate it can be certified and grounded — throughput, buffering, coherence, validation latency; not just compute.

The Shannon condition  $R < C$ , generalized from message reliability to irreversible action:

$$R_{\text{self}} < C_{\text{self}} \iff CR = \frac{R_{\text{self}}}{C_{\text{self}}} < 1, \quad \mathcal{M} = C_{\text{self}} - R_{\text{self}} \geq 0.$$

**Recoverability** is a *trajectory* property, not a snapshot — two identical states can differ in it through hidden certification debt. **Local invertibility** over  $\Delta t$  is a *mutual-information* condition  $I(\delta E; Z) \geq \iota > 0$ : distinct causes stay distinguishable, enforced by a gate  $q(t) \in [0, 1]$  with bounded  $\zeta = \tau_{\text{cert}}/\tau_{\text{upd}}$ . Minimal thermodynamic bridge — each irreversible commitment dissipates  $\geq k_B T \ln 2$  per erased bit (Landauer).

**The mathematics is a queue.** The uncertified-commitment backlog is a single-server **certification queue**;  $CR < 1$  is its stability boundary, and Shannon's  $R < C$  meets queue stability through the established theory of **effective bandwidth / effective capacity**. The contribution is the *reframing* — recoverability of irreversible action as the binding constraint, measurable from counts — not the queueing mathematics. Under acceleration the dominant failure is **premature commitment** — acting while the situation is still unresolved, merely to relieve overload.

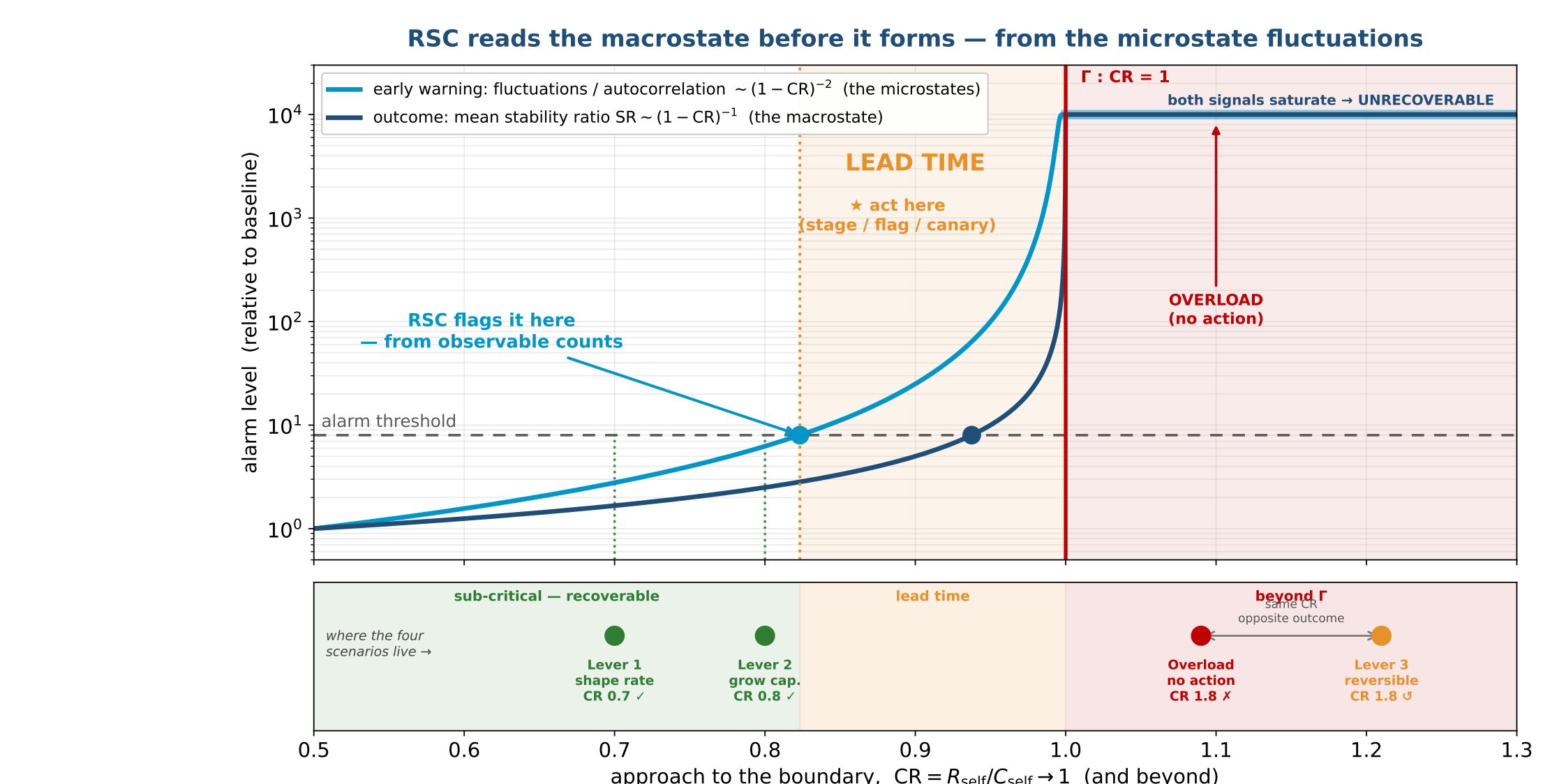
Irreversible action is admissible **only if both**:

- $\mathcal{M}(t) \geq 0$  on the relevant horizon (no sustained overload), **and**
- local invertibility holds over  $\Delta t$ .

If either fails, irreversible action causes *path-dependent contraction of future option space* that later evidence cannot undo. The admissible response is then **structural**, not “optimize harder”: **defer · repeat-first · escalate · suppress**. CR and SR are *diagnostics*, not *objectives* — making SR a target games it (Goodhart) without restoring recoverability.

## RESULTS — accuracy holds, recoverability fails

**Stability ratio**  $SR = \dot{N}_{\text{uncert}}/\dot{N}_{\text{cert}}$  — the uncertified-to-certified commitment ratio, read from counts. In the queue's decay exponent  $\theta^*$  it has the exact closed form  $SR = e^{-\theta^* \Delta t} / (1 - e^{-\theta^* \Delta t})$ , diverging as  $(1 - CR)^{-1}$  at  $\Gamma$  — the effective-bandwidth heavy-traffic limit, universal across arrival and service distributions.



**The early warning.** Fluctuations diverge *faster* than the mean —  $(1 - CR)^{-2}$  vs  $(1 - CR)^{-1}$  — so the microstate signal (cyan) crosses the alarm *before* the macrostate outcome (navy) does: that gap is the **lead time** to act (stage / flag / canary). In a simulation driven across  $\Gamma$ , **accuracy stays at 0.90 while recoverability collapses**  $0.75 \rightarrow 0.05$  — the decoupling — both diagnostics read from observable counts.

**From exception to norm.** Sustained overload ( $CR > 1$ ) was a pathological exception — overwhelm, breakdown, reached only under acute stress. AI makes it the *default*, so the imperative is to measure the *trajectory* toward  $\Gamma$  *before* the irreversible crossing.

## WHAT IT ENABLES

**Admissibility before optimization** — the hazard is not being wrong but crossing an irreversibility boundary. RSC turns recoverability into:

- an **instrument** — monitor CR, SR live from observable counts (queue depth, uncertified:certified); early warning before  $\Gamma$ ;
- a **controller** — the rate-shaping policy {defer · repeat-first · escalate · suppress}: reduce flux, grow capacity, or gate commitment;
- a complement to **uncertainty quantification** — UQ says *how uncertain*; RSC says *whether to act irreversibly* on it.

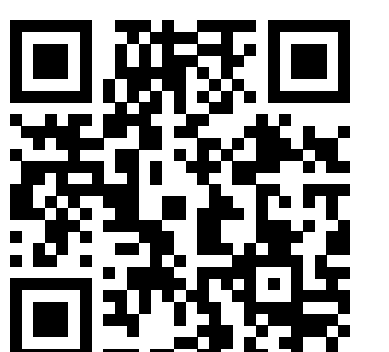
The instrument applies wherever AI compresses the loop between inference and irreversible action: a **recoverability governor** for autonomous agents; a **pacing** criterion for deployment and online learning (hold  $CR < 1$  as throughput scales); a **design principle** of option-preserving architectures. One language for humans, institutions, and AI — agency erosion under AI is a *structural* feasibility failure, not a personal one.

## FUTURE WORK & REFERENCES

**Delivered in the companion Entropy Special Issue.** The boundary law from effective-bandwidth theory; universality of the  $(1 - CR)^{-1}$  exponent; count-based estimators with an early-warning detector; the independent capacity and invertibility axes; and field signatures consistent with the prediction (honestly scoped) in GitHub and Wikipedia histories.

**Open.** Closed-loop rate-shaping control; estimating the invertibility MI from field data; a controlled regime-transition experiment.

**Preprint / priority:** van Rooyen (2026). *Entropy, Annealing, and the Continuity of Agency in Human–AI Systems*, Preprints.org doi:10.20944/preprints202601.0688. An independent treatment reaching the same thesis (Shu & Wei, arXiv:2605.01415) appeared *after* this Jan-2026 preprint.



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